Customer Shopping Behavior Analysis

Uncovering insights from 3,900 purchases to guide strategic business decisions

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Dataset Overview

3,900

Purchase Records

Transactional data analyzed

18

Data Columns

Customer and purchase details

37

Missing Values

In Review Rating column

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Data Preparation Process

01

Data Loading

Imported dataset using pandas

02

Initial Exploration

Checked structure and summary statistics

03

Missing Data Handling

Imputed Review Rating using median by category

04

Feature Engineering

Created Age Groups and Purchase Frequency Days

05

Database Integration

Loaded cleaned data into PostgreSQL

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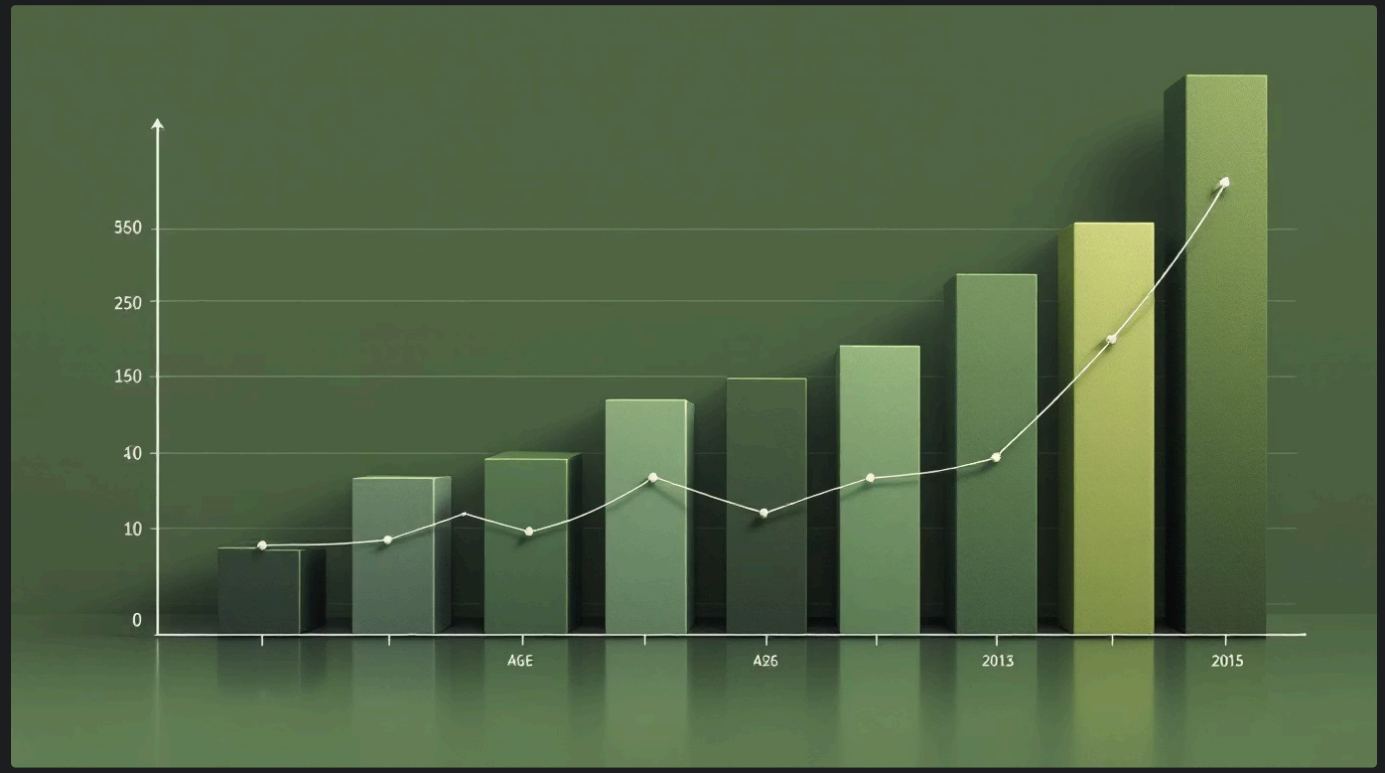
Revenue by Age Analysis



Top Revenue Generators

- Age 49: $5,552
- Age 69: $5,484
- Age 25: $5,372
- Age 41: $5,282

Age diversity shows broad customer base

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Discount Strategy Insights

High-Spending Discount Users

839 customers used discounts while spending above average

Discount-Dependent Products

Hats (50%), Sneakers (49.66%), Coats (49.07%) most discounted

Policy Balance

Review discount policy to balance sales with margin control

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Product Performance

Top-Rated Products

1. Gloves: 3.86 rating
2. Sandals: 3.84 rating
3. Boots: 3.82 rating
4. Hat: 3.80 rating
5. Skirt: 3.79 rating

Shipping Comparison

Standard: $58.46 avg

Express: $60.48 avg

Express shipping correlates with higher purchase amounts

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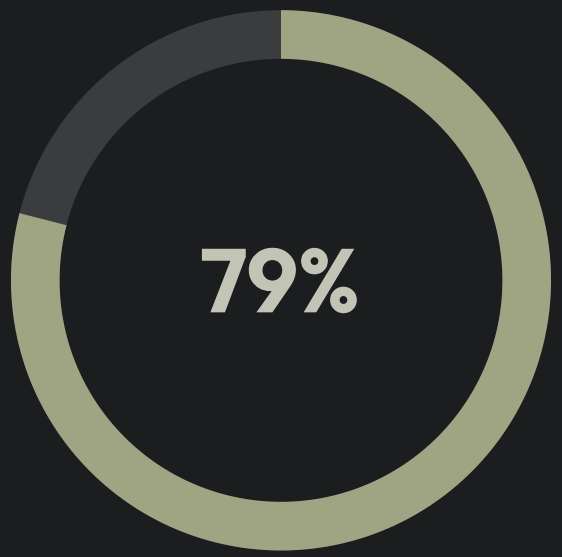
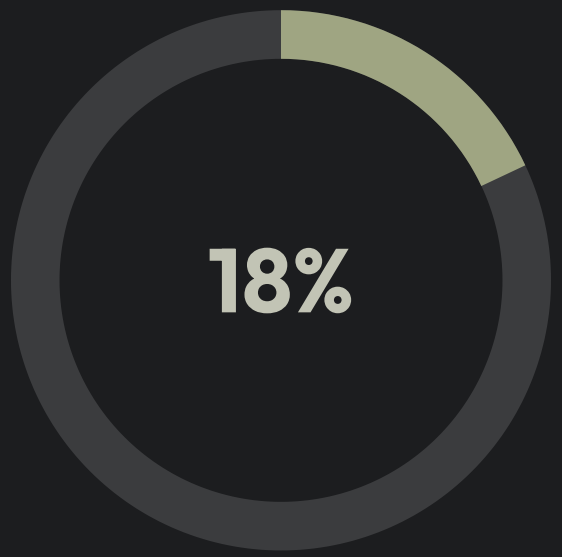
Customer Segmentation



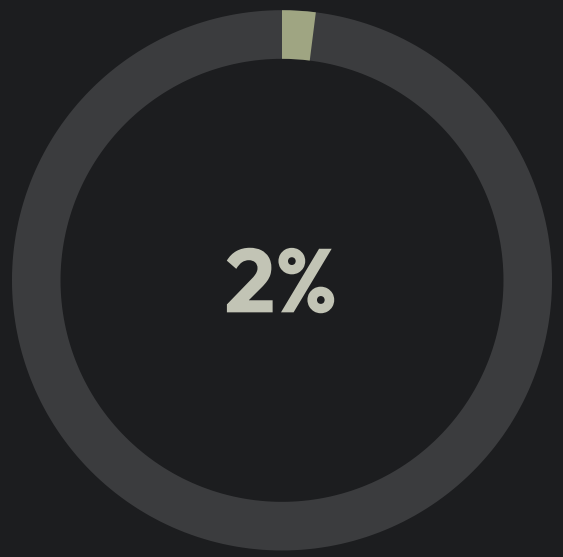
Loyal Customers

3,116 customers with consistent purchase history



Returning

701 customers making repeat purchases



New

83 first-time customers

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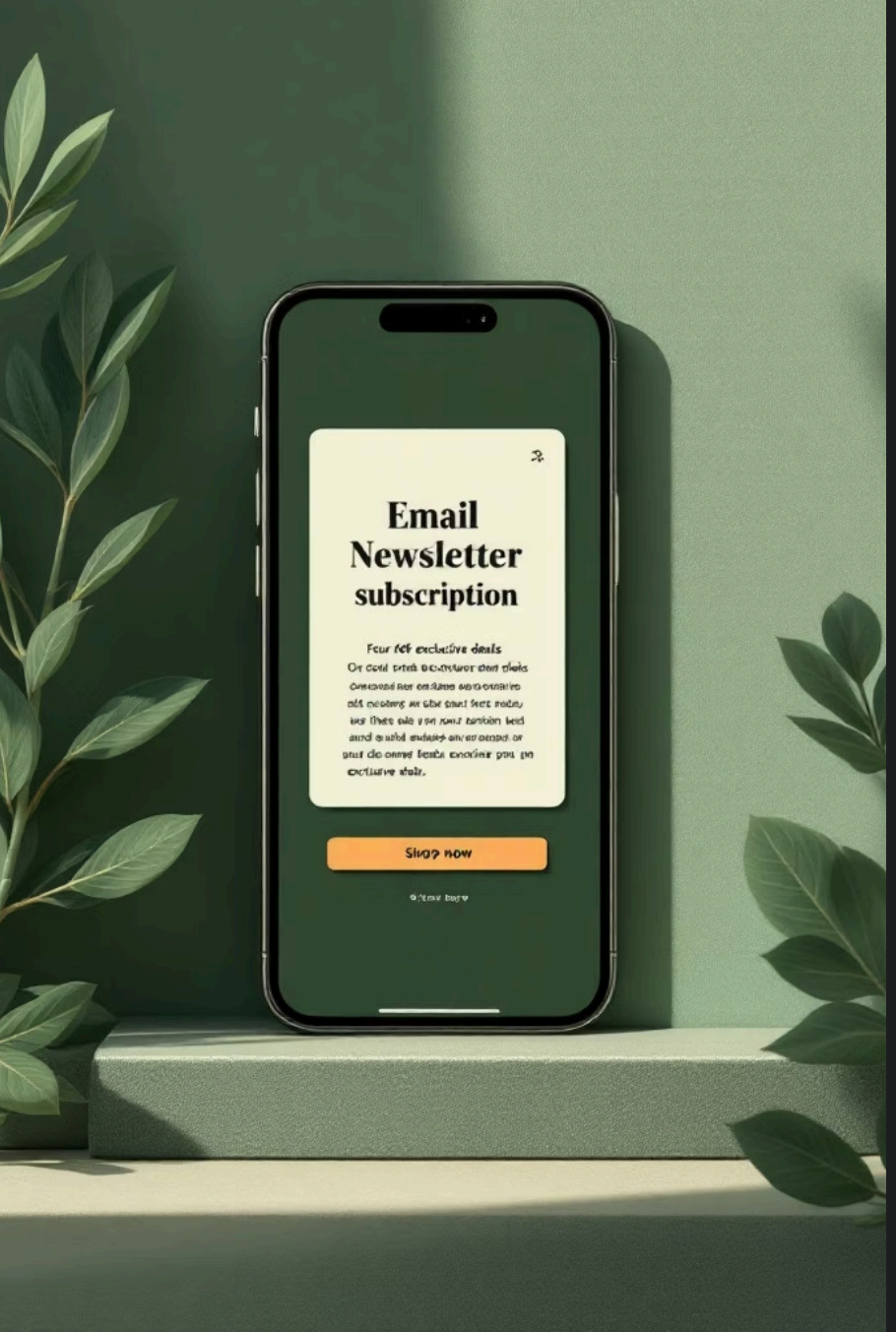

Subscription Impact

Non-Subscribers

2,847 customers

$59.87 average spend

$170,436 total revenue

Subscribers

1,053 customers

$59.49 average spend

$62,645 total revenue

Repeat buyers (over 5 purchases) show higher subscription likelihood

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Power BI Dashboard

Interactive Filters

Subscription, Gender, Category, Shipping Type

Key Metrics

Customers, Average Purchase, Review Rating

Visualizations

Revenue by Category, Sales by Age, Customer Segments

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Strategic Recommendations

Boost Subscriptions

Promote exclusive benefits

Loyalty Programs

Reward repeat buyers

Targeted Marketing

Focus on high-revenue age groups

Product Positioning

Highlight top-rated items

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